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WHEN DO FUTURES DESTABILIZE SPOT PRICES?*

BY DAVID M. NEWBERY¹

1. INTRODUCTION

Do speculators, trading on futures markets, tend to stabilize or destabilize spot prices? The question, though old, has a renewed urgency, as recent technological advances in communications and computing have led to a rapid growth in the number of futures markets and in their traded volume. If these developments increase the underlying price instability, then this may have an adverse effect on activities which lack access to the hedging functions of futures markets. In particular, although financial institutions may be able to insure against short term movements in exchange rates, interest rates, and asset prices, firms, especially small unquoted companies, may be unable to adequately insure their long term physical investments. In such cases, greater instability may have real economic costs.

Friedman's argument [1953] was that speculators make profits by moving prices towards the correct, i.e. equilibrium, level and in so doing, tend to reduce the fluctuations of the futures price around the best estimate of the future spot price. Hart [1977] showed that this argument was unconvincing, for in his model, a sophisticated speculator can make money by exploiting the naive forecasting rules of less sophisticated agents, and profitably destabilizing the futures price. Newbery [1984] also demonstrated that if a producer has market power, then even if all agents have rational expectations, it may pay the dominant producer to destabilize the spot market, and possibly to engage in destabilizing speculation.

If one wishes to argue for the stabilizing effects of speculation, then, it is important to exclude irrational forecasting behaviour and market power. But does it follow that if all agents have common knowledge, rational expectations, and no market power, speculation will be stabilizing? Even if one rules out 'speculative bubbles' (Tirole, [1985]), the theoretical answer must be no, for Kawai [1983] has constructed a rational expectations model of storable commodities in which the introduction of a futures market, which facilitates speculation, may destabilize the spot market. Turnovsky and Campbell [1985] argue in contrast that in their rational expectations model, futures markets always improve the stability of the spot market. In so doing, they support the conclusion reached by earlier, less satisfactory, theoretical investigations.¹

The empirical evidence, surveyed by Cox [1976], whilst not completely unam-

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¹ The papers cited include Working [1958], Stein [1961], Peck [1976] and Turnovsky [1979].

biguous, supports the view that futures markets do not increase spot price instability, and often reduce instability. Cox argues that the theoretical reason for the improved stability is that the futures market increases the information available to traders. Whilst this may be the most important function of futures markets, it is not considered further here, since the discussion assumes no change in information consequent upon opening a futures market.

The purpose of this note is to explain how speculation may destabilize prices, and to provide a simple example to illustrate the point. Turnovsky and Campbell [1985] point out, and Kawai [1983] also demonstrates, that models with storage are analytically intractable, and are in any case unnecessary to illustrate the point.

The basic idea is very simple. Futures markets offer insurance, and thereby encourage agents to make riskier decisions than they otherwise would. In one-commodity partial equilibrium models, the effect is typically to increase the supply of the risky crop, and hence, to lower the average market clearing price. If, however, risk is additive and supply and demand schedules are linear (as in the studies cited), this average supply response has no direct effect on the variability of supply, and hence, no direct effect on the price variability, measured by its variance. (It will typically increase its coefficient of variation slightly by reducing the mean price).

Instead, futures markets affect spot price variability by increasing the amount of storage, as it reduces the risk of holding stocks, and hence effectively lowers the cost of storage. In Turnovsky and Campbell's model, storage in non-stochastic, so increased storage necessarily reduces the average degree of interyear price variability. In Kawai's model, storage is subject to (unexplained) stochastic shocks, and more storage means that these shocks will be larger, and, if they dominate the other random shocks, may destabilize the spot price.

Thus, futures markets encourage risk taking, and whether this stabilizes or destabilizes the spot price depends on whether the risky activity tends to stabilize or destabilize spot prices. Normally storage is thought to be relatively non-stochastic but subject to price risk, and if it is encouraged, then price stability will also be encouraged. But once it is recognised that (on the assumption of rational expectations) the main effect futures markets have is on risk taking, then it is easy to see how risk taking might destabilize spot markets. The simplest model is one in which producers can produce a non-storable good in either a safe way or a risky way. (Think of maize or rice in which traditional varieties are resistant to disease, but less productive on average than new varieties, which, however, are more prone to drought and disease). The risky method is on average more productive, so farmers will choose to allocate a fraction of their land to the risky method. A futures market will reduce the risk of producing the risky crop, and hence increase the fraction of land allocated to it. This will increase the output variability and, as there is no storage, the increased output variability will increase the variability of the spot price.

The next section provides a simple example to demonstrate that as speculators increase their level of activity on futures markets, so they provide increasing levels

of insurance to farmers, who are induced to increase their supply of the risky crop, and so increase the spot price variability. The example has been kept simple by having no storage, only producers and speculators trading in futures markets, and making standard linearity assumptions that permit closed form solutions (as in Newbery and Stiglitz, [1981], Kawai, [1983], and Turnovsky and Campbell, [1985]).

2. THE MODEL

There are n identical farmers, each with $1/n$ units of land ('acres'). There are two methods of producing output. The safe method yields 1 unit of output/acre with certainty, the risky method yields $1 + \tilde{u}$, a normally distributed random variable whose expected value exceeds that of the safe crop:

$$(1) \quad E(1 + \tilde{u}) = 1 + r, \quad r > 0; \quad \text{Var}(1 + \tilde{u}) = \text{Var} \tilde{u} = \sigma^2.$$

Farmers must choose the fraction of land, x , to allocate to the risky crop at the start of the year, before the realisation of u is observed. Total output per acre for the representative farmer who sets $x = x^*$ is

$$(2) \quad \tilde{q} = 1 - x^* + x^*(1 + \tilde{u}) = 1 + x^*\tilde{u}.$$

The market clearing price after the harvest, $\tilde{p}$, is a random variable given by the linear demand schedule

$$(3) \quad \tilde{p} = 1 + \frac{1}{\varepsilon} - \frac{1}{\varepsilon} \tilde{q}.$$

(Thus units are chosen so that if farmers specialise in the safe crop the average price will be unity, as will be output, and the elasticity of demand at this average price is ε). Since u is normally distributed, so will be the price, p .

Farmers have rational expectations and know the joint distribution of p and q . In equilibrium each farmer knows the total fraction of land allocated to the risky crop, x^* , and believes that the price depends on the state of the world, u , according to

$$(4) \quad p = p\{q(x^*, u)\} = 1 - x^*u/\varepsilon.$$

The optimal choice of x given $p(u)$ is then x^* , completing the description of the rational expectations equilibrium.

3. FUTURES MARKET

A futures market offers farmers the chance to sell futures at a known price, f , at the start of the crop year, when they are choosing their cropping pattern, x , and before they know the state of the world, and hence the price of their output, p . Farmers sell futures to speculators, who are distinguished by their lack of income from production (or from any other source correlated with u). If a farmer

sells z/n futures his income is

$$(5) \quad \tilde{y} = \frac{1}{n} \{fz + \tilde{p}(\tilde{q} - z)\} = \frac{1}{n} \{fz + \tilde{p}(1 - z + x\tilde{u})\}.$$

There are N identical speculators, and if a speculator sells z^s/N futures, his income will be

$$(6) \quad \tilde{y}^s = \frac{1}{N} (f - \tilde{p})z^s.$$

Farmers and speculators make their choices (of x , z , z^s) at the start of the crop year before u is known in order to maximize the expected utility of income. They face normally distributed risks, and if they have exponential utility functions (i.e. constant absolute risk aversion, k), this is equivalent to maximizing

$$(7) \quad W = Ey - \frac{1}{2} k \text{Var } y$$

(where E is the expectations operator) as in the standard Capital Asset Pricing Model. Alternatively, (7) can be viewed a second order approximation to a general utility function. (See Newbery and Stiglitz, [1981]). Farmers will therefore choose x and z to maximize W , where

$$(8) \quad nW = fz + (1-z)Ep + xEpu - \frac{1}{2} \frac{k}{n} \{(1-z)^2 \text{Var } p + 2(1-z)x \text{Cov}(p, pu) + x^2 \text{Var } pu\}.$$

The chosen values for x , z can be found from the first order conditions:²

$$(9) \quad x = \frac{\frac{n}{k} Epu - (1-z) \text{Cov}(p, pu)}{\text{Var } pu},$$

$$(10) \quad z = 1 + \frac{\frac{n}{k} (f - Ep) + x \text{Cov}(p, pu)}{\text{Var } p}.$$

(If there is no futures market, then $z=0$ in (9)). Equation (10) gives the futures sales by farmers. Speculators, on the other hand, have no income from farming, and choose their future sales to maximise expected utility. If the N identical speculators have coefficients of absolute risk aversion K , their income is given by (6), and their choice of z^s (possibly negative) will be given by

$$(11) \quad z^s = \frac{\frac{N}{K} (f - Ep)}{\text{Var } p}.$$

² Interior solutions for x , $0 < x < 1$, are assumed, but will require sufficiently low values of r (the excess output of the risky crop) relative to its risk (measured by σ^2) and/or sufficiently high values of risk aversion, k . Otherwise $x=1$, equation (9) will be an inequality, and futures markets will not allow any increased risk bearing.

The futures price f must be such that the futures market clears, or total futures sales are zero:

$$(12) \quad z + z^s = 0 = 1 + \frac{x \operatorname{Cov}(p, pu)}{\operatorname{Var} p} + \frac{\left[\frac{n}{k} + \frac{N}{K}\right](f - Ep)}{\operatorname{Var} p}.$$

Substitute this in equation (10) to solve for z :

$$(13) \quad z = \beta \left[1 + \frac{x \operatorname{Cov}(p, pu)}{\operatorname{Var} p} \right].$$

where

$$(14) \quad \beta = \frac{Nk}{Nk + nK}, \quad 0 \leq \beta \leq 1.$$

If speculators are infinitely risk averse ($K \rightarrow \infty$), then $\beta = 0$ and there will be no active trading on the futures market, whereas if speculators are risk neutral, $K = 0$, and $\beta = 1$. Thus β gives a measure of the extent of speculative activity in the economy. Substitute (13) into (9) to obtain

$$(15) \quad x(1 - \beta\rho^2) = \frac{\frac{n}{k}Epu - (1 - \beta)\operatorname{Cov}(p, pu)}{\operatorname{Var} pu}$$

where

$$\rho = \frac{\operatorname{Cov}(p, pu)}{\sqrt{\operatorname{Var} p \operatorname{Var} pu}}$$

is the correlation coefficient between p and pu . Equation (15) is a general equation, holding for any bivariate normally distributed variables p , u , and can now be solved for the particular simple case of pure supply risk. The correlation coefficient $\rho = -1$ (to a high degree of approximation) and the values of the other expressions are as follows

$$Epu = r - x^*(r^2 + \sigma^2)/\varepsilon$$

$$\operatorname{Var} pu = (1 - 2rx^*/\varepsilon)^2\sigma^2 + o(\sigma^4), \quad (o(\sigma^4) = 2(x^*/\varepsilon)^2\sigma^4)$$

$$\operatorname{Var} p = (x^*/\varepsilon)^2\sigma^2$$

$$\operatorname{Cov}(p, pu) = -(x^*/\varepsilon)(1 - 2rx^*/\varepsilon)\sigma^2$$

The term n/k can be interpreted by noting that if farmers set $x = 0$ their income is $1/n$, and their coefficient of relative risk aversion k_y will then be $k/n = R$, a dimensionless constant. Equation (15) can be written as

$$(16) \quad x(1 - \beta) = \frac{r\varepsilon^2 - \varepsilon x^*(r^2 + \sigma^2)}{R(\varepsilon - 2rx^*)^2\sigma^2} + \frac{(1 - \beta)x^*}{(\varepsilon - 2rx^*)}.$$

This can be solved for $x = x^*$, given values for r , R , σ^2 , ε , and β . The claim

advanced here is that $dx^*/d\beta$ is positive for some values of these parameters, and the simplest way to demonstrate this is to examine small perturbations around $\beta=0$, for small values of r . It is easy to demonstrate that $x^*\rightarrow 0$ as $r\rightarrow 0$, i.e. as the returns to risk bearing go to zero, so does the risk undertaken. It is, however, necessary to check that the equilibrium for $x\rightarrow x^*$ is stable, since it is typically quite usual to find paradoxical results for unstable equilibrium. For stability at $\beta=0$, $r\rightarrow 0$, we require $|\partial x/\partial x^*| < 1$, or, from (16):

$$(17) \quad \left| \frac{\partial x}{\partial x^*} \right| = \left| -\frac{1}{\varepsilon R} + \frac{1}{\varepsilon} \right| = \left| \frac{1}{\varepsilon} \left(1 - \frac{1}{R} \right) \right| < 1.$$

Now differentiate (16) totally with respect to x at $x=x^*$, $\beta=0$, $r\rightarrow 0$:

$$(18) \quad 1 - x\beta' + \frac{x\beta'}{\varepsilon} = \frac{\partial x}{\partial x^*}$$

where $\partial x/\partial x^*$ is the derivative of the right side of (16) with respect to x^* , and β' is $d\beta/dx$, whose sign we wish to examine. From the stability condition (17), equation (18) can be rearranged to give

$$x\beta' \left(1 - \frac{1}{\varepsilon} \right) = 1 - \frac{\partial x}{\partial x^*} > 0$$

so $dx/d\beta = 1/\beta'$ is positive if $\varepsilon > 1$.

The variance of prices (and its coefficient of variation) in this model are both increasing in x^* , so if demand is elastic, the effect of gradually introducing a futures market is to increase the degree of risk taking by farmers and hence to increase price instability (at least, if the returns to risk bearing, r , are small).

A numerical example to confirm this result has parameters $\varepsilon=2$, $R=1$, $\sigma=0.3$, $r=0.03$. At $\beta=0$ (no futures market) $x^*=0.335$, whilst with $\beta=0.5$, x^* increases to 0.443. (These figures are calculated using the exact expression for $\text{Var } pu$, adding the term $2(x^*/\varepsilon)^2\sigma^4$. The correlation coefficient $\rho=-0.975$). The coefficient of variation of price increases from 5 percent to 6.6 percent or by one third.

4. CONCLUSION

Futures markets offer speculators a chance to assume price risks in the expectation of making profits. If all agents are equally well informed and hold rational expectations, and if no agents have market power, then speculators cannot manipulate or mislead — two obvious reasons why their activities might be destabilizing. In such cases speculators affect stability by offering price insurance which reduces the risks of certain activities, and stimulates their supply. Often these risky activities tend to reduce price instability — storage is the leading example — but clearly if the risky activity increases price risk, then speculators will tend to increase price instability. In the paper this was demonstrated for a case where producers were encouraged to change to a more risky but more profitable mode

of production.

If futures markets extend only for a year or less, then agents making long term investment decisions may face greater risk as a result of the increased volume of short term speculative activity. It is possible that the costs of this greater long term risk will exceed the benefits to the short term hedges of providing the futures market.

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